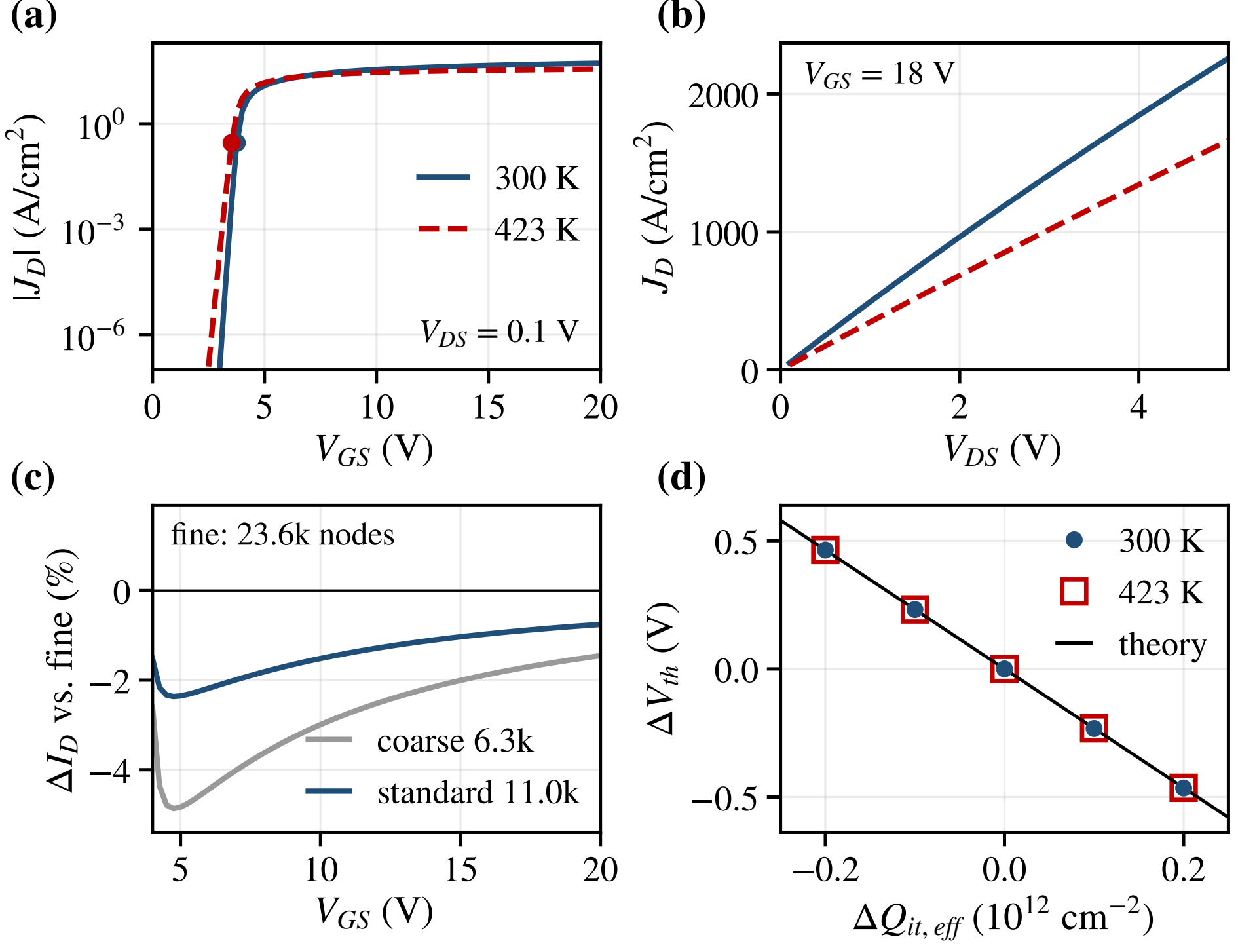
(a)
$|I_D|$ (A/cm^2)
10^0
10^{-3}
10^{-6}
300 K
423 K
$V_{DS} = 0.1$ V
0
5
10
15
20
V_{GS} (V)
(b)
$V_{GS} = 18$ V
J_D (A/cm^2)
2000
1000
0
0
2
4
V_{DS} (V)
(c)
fine: 23.6k nodes
ΔI_D vs. fine (%)
0
−2
−4
coarse 6.3k
standard 11.0k
5
10
15
20
V_{GS} (V)
(d)
ΔV_{th} (V)
0.5
0.0
−0.5
300 K
423 K
theory
−0.2
0.0
0.2
$\Delta Q_{it,\,eff}$ (10^{12} cm^{-2})